AI Assisted Coding

Week7 - Monday

Name: Lalu prasad Aroori

Roll No.: 2303A51948

Batch No.: 07

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Week 7 – Monday

- **Task 1 – Portfolio Website Design**

---

1) Prompt to Give in VS Code

```
Create a responsive personal portfolio website using HTML, CSS, and
JavaScript.

Requirements:
- Sections: About Me, Projects, Contact
- Use Flexbox or CSS Grid for layout
- Add smooth scrolling navigation
- Suggest a modern color palette and typography
- Make the design responsive for mobile devices
```

---

2) Code

HTML

```
<!DOCTYPE html>
<html>
<head>
<title>My Portfolio</title>
<link rel="stylesheet" href="style.css">
```

```
</head>

<body>

<nav>
<a href="#about">About</a>
<a href="#projects">Projects</a>
<a href="#contact">Contact</a>
</nav>

<section id="about">
<h1>About Me</h1>
<p>I am a web developer passionate about AI-assisted coding.</p>
</section>

<section id="projects">
<h1>Projects</h1>
<p>Portfolio Website, Resume Builder, Inventory System</p>
</section>

<section id="contact">
<h1>Contact</h1>
<p>Email: example@email.com</p>
</section>

<script src="script.js"></script>
</body>
</html>
```

CSS

```
body{
font-family: Arial;
margin:0;
background:#f5f7fa;
}

nav{
display:flex;
justify-content:center;
background:#2c3e50;
}
```

```
nav a{
color:white;
padding:15px;
text-decoration:none;
}

section{
padding:60px;
text-align:center;
}
```

JavaScript (Smooth Scroll)

```
document.querySelectorAll('nav a').forEach(link => {
link.addEventListener('click', function(e){
e.preventDefault()
document.querySelector(this.getAttribute('href')).scrollIntoView({
behavior:'smooth'
})
})
})
```

---

3) Output

Webpage containing:

- Navigation bar
- About Me section
- Projects section
- Contact section
- Smooth scrolling navigation

---

4) Remarks

✓ AI suggested **modern color palette**
✓ Flexbox used for responsive navigation
✓ Smooth scrolling improves UX
✓ Clean portfolio structure

🔹 **Task 2 – Online Store Product Page**

---

1) Prompt

Create a responsive product display page for an online store.

Requirements:
- Product image, title, price
- "Add to Cart" button
- Use BEM CSS methodology
- Add hover effects
- Show alert when clicking Add to Cart

---

2) Code

HTML

```
<div class="product">
<img class="product__image" src="product.jpg">

<h2 class="product__title">Wireless Headphones</h2>

<p class="product__price">$99</p>

<button class="product__button" onclick="addCart()">
Add to Cart
</button>
</div>
```

CSS

```
.product{
width:300px;
margin:auto;
text-align:center;
border:1px solid #ddd;
padding:20px;
}
```

```
.product__image{
width:100%;
}

.product__button{
background:#27ae60;
color:white;
border:none;
padding:10px;
cursor:pointer;
}

.product__button:hover{
background:#2ecc71;
}
```

JavaScript

```
function addCart(){
alert("Product added to cart!")
}
```

---

3) Output

Product page displaying:

- Product image
- Product name
- Price
- Add to Cart button

Clicking button → Alert appears.

---

4) Remarks

✓ BEM methodology improves maintainability
✓ Hover effects improve UX
✓ Responsive layout supported

◆ Task 3 – Event Registration Form

---

1) Prompt

Create an accessible event registration form.

Requirements:
- Fields: Name, Email, Phone, Session selection
- Add JavaScript validation
- Use labels and ARIA attributes
- Style professionally with CSS

---

2) Code

HTML

```
<form id="form">

<label>Name</label>
<input type="text" id="name" required>

<label>Email</label>
<input type="email" id="email" required>

<label>Phone</label>
<input type="tel" id="phone">

<label>Session</label>
<select>
<option>AI Workshop</option>
<option>Web Development</option>
</select>

<button type="submit">Register</button>

</form>
```

JavaScript

```
document.getElementById("form").onsubmit=function(e){

let name=document.getElementById("name").value

if(name.length<3){
alert("Name must be at least 3 characters")
e.preventDefault()
}

}
```

---

3) Output

Registration form with:

- Name input
- Email input
- Phone input
- Session selection

Validation shows alerts if incorrect.

---

4) Remarks

✓ Accessible form with labels
✓ Validation improves data accuracy
✓ Professional layout

🔹 Task 4 – Resume Website

---

1) Prompt

Create a one-page resume website.

Sections:
- Profile Summary
- Skills
- Education

- Experience

Use CSS Grid for layout and smooth navigation.

---

2) Code

HTML

```
<section id="profile">
<h2>Profile Summary</h2>
<p>Computer Science Student interested in AI.</p>
</section>

<section id="skills">
<h2>Skills</h2>
<p>Python, Java, HTML, CSS</p>
</section>

<section id="education">
<h2>Education</h2>
<p>SR University</p>
</section>

<section id="experience">
<h2>Experience</h2>
<p>Cloud Computing Internship</p>
</section>
```

CSS

```
body{
display:grid;
grid-template-columns:1fr;
gap:20px;
padding:30px;
font-family:Georgia;
}

section{
border-bottom:1px solid #ccc;
}
```

---

3) Output

Single-page resume website displaying:

• Profile
• Skills
• Education
• Experience

---

4) Remarks

✓ CSS Grid ensures structured layout
✓ Professional typography
✓ Simple and readable design

🔹 Task 5 – Restaurant Menu Page

---

1) Prompt

Design a restaurant menu page.

Requirements:
- Menu items with name, description, price
- Card layout
- Food-friendly color palette
- Hover effects
- Responsive design

---

2) Code

HTML

```
<div class="menu">

<div class="card">
<h3>Pizza</h3>
<p>Cheese loaded pizza</p>
<span>$12</span>
```

```
</div>

<div class="card">
<h3>Burger</h3>
<p>Grilled chicken burger</p>
<span>$8</span>
</div>

</div>
```

CSS

```
body{
background:#fff8e1;
font-family:Arial;
}

.menu{
display:flex;
gap:20px;
}

.card{
background:white;
padding:20px;
border-radius:8px;
box-shadow:0 0 10px rgba(0,0,0,0.1);
}

.card:hover{
transform:scale(1.05);
}
```

---

3) Output

Restaurant menu displaying:

- Food cards
- Item name
- Description
- Price

Cards enlarge on hover.

---

4) Remarks

✓ Warm color palette improves food presentation
✓ Card layout enhances readability
✓ Hover effects increase interactivity

Final Conclusion for Lab 14

AI-Assisted Web Development Benefits:

✓ Faster UI generation
✓ Better layout suggestions
✓ Improved UX ideas
✓ Responsive design recommendations